

Characterization of methane emissions from rice fields in Asia.

III. Mitigation options and future research needs

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Abstract

Methane (CH₄) emissions from rice fields were determined using automated measurement systems in China, India, Indonesia, Thailand, and the Philippines. Mitigation options were assessed separately for different baseline practices of irrigated rice, rainfed, and deepwater rice. Irrigated rice is the largest source of CH₄ and also offers the most options to modify crop management for reducing these emissions. Optimizing irrigation patterns by additional drainage periods in the field or an early timing of midseason drainage accounted for 7–80% of CH₄ emissions of the respective baseline practice. In baseline practices with high organic amendments, use of compost (58–63%), biogas residues (10–16%), and direct wet seeding (16–22%) should be considered mitigation options. In baseline practices using prilled urea as sole N source, use of ammonium sulfate could reduce CH₄ emission by 10–67%. In all rice ecosystems, CH₄ emissions can be reduced by fallow incorporation (11%) and mulching (11%) of rice straw as well as addition of phosphogypsum (9–73%). However, in rainfed and deepwater rice, mitigation options are very limited in both number and potential gains. The assessment of these crop management options includes their total factor productivity and possible adverse effects. Due to higher nitrous oxide (N₂O) emissions, changes in water regime are only recommended for rice systems with high baseline emissions of CH₄. Key objectives of future research are identifying and characterizing high-emitting rice systems, developing site-specific technology packages, ascertaining synergies with productivity, and accounting for N₂O emissions.

Introduction

There is an increasing pressure on the rice-growing resources, especially in Asia where more than 90% of rice is grown and consumed (Blake, 1992; Becker, 1993). Rice cultivation over thousands of years has sustained Asian population, which is currently growing at 1.8% a year. Wetland rice fields, however, release the greenhouse gas methane (CH₄) and thus, affect the radiative budget of the earth (Minami & Neue, 1994; Neue & Sass, 1999). Given the expected doubling in rice production in Asia, conducting research that will help developing countries grow more rice on limited land, in ways that do not harm the environment and that benefit both farmers and consumers, will be

critical. There is a need to evaluate the interaction between climate change and rice production to provide a sound basis for future decisions and technology developments by policymakers, agriculturists, environmentalists, rice producers, and rice consumers alike.

Global climate change has been recognized as a major threat for future development in the United Nations Framework Convention on Climate Change (UNFCCC) in 1992. The ultimate goal of the convention is the stabilization of greenhouse gas concentration in the atmosphere at a level that would prevent anthropogenic interference with the climate system. Before target stabilization can be specified, a national greenhouse gas inventory is necessary for each country to formulate mitigation policies for international agreements. On

the other hand, large portions of Asia's rice fields are located on lowlands that were natural wetlands before agricultural use. These areas have already been a source of CH₄ in their pristine state. The introduction of rice substantially enhanced the turnover of organic material and, in the next step, CH₄ emissions.

The Interregional Research Program on Methane Emissions from Rice Fields has established a network of measurement stations in China, India, Indonesia, Thailand, and the Philippines. This work was initiated by the International Rice Research Institute, Philippines, in collaboration with national agricultural research institutes and the Fraunhofer Institute for Atmospheric Environmental Research, Germany, to quantify emissions from major rice-growing systems and to identify possible strategies for mitigation. Generic strategies on mitigating CH₄ emissions have been formulated recently (Lindau et al., 1993; Wassmann et al., 1993; Neue et al., 1995; Ranganathan et al., 1995; Shin et al., 1996; Yagi et al., 1997; Minami, 1997), but information on their feasibility and efficiency in different rice-growing environments is still lacking. The immense variability in environmental and management factors in the 144 million ha of annually harvested rice fields (IRRI, 1993) demands site-specific assessments on mitigating emissions.

Flux measurements at the eight sites of this network showed a large variability in CH₄ emissions addressed in this series of articles. The first volume of this series (Wassmann et al., this issue, b) dealt with site-to-site variation under a given crop management. The second volume focused on differences among irrigated, rainfed, and deepwater rice (Wassmann et al., this issue, c). In this third volume, we evaluate crop management impacts with emphasis on possible mitigation options. Moreover, we assessed the findings of this network in the context of future research needs.

Materials and methods

Methane measurements were conducted with a standard automated closed chamber system designed and manufactured by the Fraunhofer Institute for Atmospheric Environmental Research (Wassmann et al., this issue, b). The design of the system is suitable for continuous CH₄ flux measurements in the different rice ecosystems over long time periods. Each station in the network was supplied with 12 closed chambers, so that four treatments could be investigated with three replicates of each.

The database used in this study comprises only seven stations:

- Beijing, China (Z. Wang et al., this issue)
- Hangzhou, China (Lu et al., this issue)
- Maligaya, Philippines (Corton et al., this issue)
- Cuttack, India (Adhya et al., this issue)
- Jakenan, Indonesia (Setyanto et al., this issue)
- Los Baños, Philippines (Wassmann et al., this issue, a; Abao et al., this issue)
- Prachinburi, Thailand (Chareonsilp et al., this issue)

These stations were located in principal rice-growing areas in Asia and represented irrigated rice (Beijing, Hangzhou, and Maligaya), rainfed rice (Jakenan, Cuttack, and Los Baños) and deepwater rice (Prachinburi). Another station of this network (New Delhi) did not yield clues for mitigating emissions because baseline emissions were already very low (Jain et al., this issue; see Figure 1 for comparison). The rainfed and deepwater stations included irrigated rice for comparative purposes. Measurements were started in 1993 and ended in 1998, covering up to 10 consecutive seasons per station.

Results and discussion

Rice is grown in very diverse environments, applying a wide range of crop management practices as can be seen in GIS data in Knox et al. (this issue). Mitigation strategies can only be defined for specific target systems. According to ecosystem and crop management, five different baseline practices were distinguished as follows:

Irrigated	I ₁ (continuous flooding, organic amendments)
	I ₂ (midseason drainage, organic amendments)
	I ₃ (continuous flooding, no organic amendments)
Rainfed:	R
Deepwater:	D

Irrigated rice accounts for 51% of the harvest rice area in Asia. For this ecosystem, water regime and organic amendments are the major determinants for the magnitude of CH₄ emissions (Wassmann et al., this issue, c). Continuous flooding and fresh organic manure resulted in highest emissions, whereas emissions were reduced in some cases by several orders of magnitude—through temporary soil aeration and omission of organic manure. Other management components such as type

of inorganic fertilizer, rice cultivar, etc. had a modulating effect on emissions (i.e., they affected emissions within a range that was set by water management and organic amendment) (Wassmann et al., this issue, c). Rainfed (27% of the harvest rice area) and deepwater rice (10% of the harvest rice area) offer very limited management options and are not further split into different baseline practices.

The experiments of the network encompassed simultaneous records of different crop management options. From this database, we have extracted pairs of two management options, i.e., one representing a baseline practice (I_1 , I_2 , I_3 , R, or D) and the other a distinct modification of this management practice termed as “mitigation practice.” Figure 1 shows seasonal emissions recorded for the respective baseline practice. All

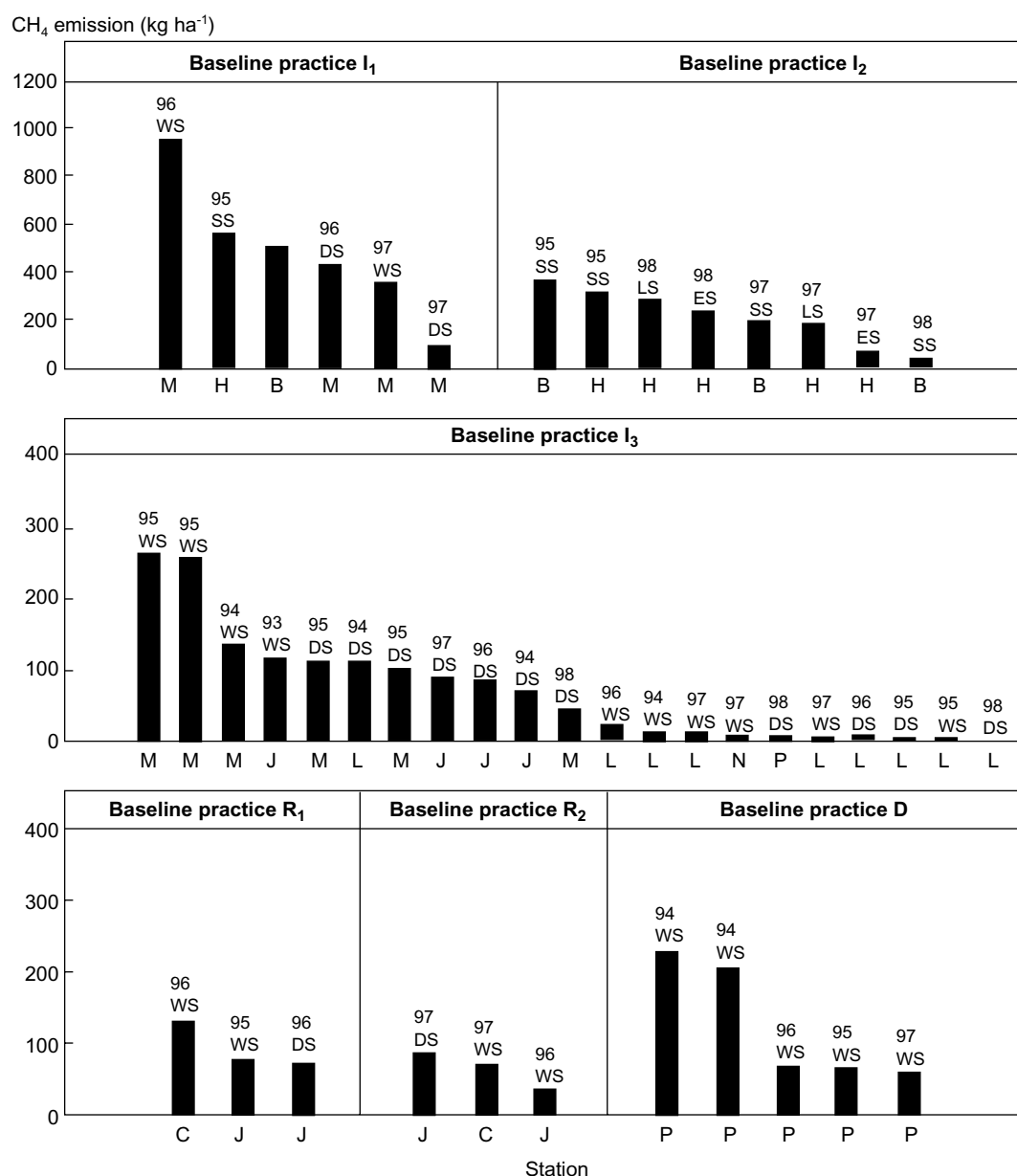


Figure 1. Seasonal CH₄ emissions of experiments with baseline practices (see text for explanation of baseline practice). Year and season of the experiment are given in labels. First letter of Hangzhou, Beijing, Maligaya, Jakenan, Los Baños, New Delhi, Cuttack, and Prachinburi indicate experimental station

Table 1. Mitigation practices for baseline practice I_1 (continuous flooding, organic amendments): seasonal emission refers to mitigation practice; reduction effect refers to emission from mitigation practice as compared with emission from baseline practice; seasons are specified as dry (DS), wet (WS), early (ES), late (LS), and single season (SS)

Mitigation practice	Seasonal emission (kg ha ⁻¹)	Reduction effect		Experiment		Yield impact ^a (%)	Remark
		Net reduction (kg ha ⁻¹)	Relative reduction ^s (%)	Station	Season		
Midseason drainage	385	118	23**	Beijing	1995 SS	14*	
	312	245	44 ns	Hangzhou	1995 SS	-1 ns	
	51	38	43**	Maligaya	1997 DS	-2 ns	
	323	25	7 ns	Maligaya	1997 WS	2 ns	
Alternate flooding/drainage	216	341	61**	Hangzhou	1995 SS	0 ns	
	207	296	59**	Beijing	1995 SS	12*	
Rice straw compost	178	242	58**	Maligaya	1996 DS	4 ns	Baseline practice: fresh rice straw
Phosphogypsum (3 t ha ⁻¹)	353	599	63**	Maligaya	1996 WS	2 ns	
	308	112	27**	Maligaya	1996 DS	1 ns	
	599	353	37**	Maligaya	1996 WS	1 ns	
Direct wet seeding	272	76	22**	Maligaya	1997 WS	-28*	Baseline practice: transplanting
Direct wet seeding and midseason drainage	75	14	16**	Maligaya	1997 DS	-15*	
	48	41	46**	Maligaya	1997 DS	-19*	
	150	198	57**	Maligaya	1997 WS	-36*	
Midseason drainage and no organic matter	26	477	95**	Beijing	1995 SS	-4 ns	
	239	318	57**	Hangzhou	1995 SS	-3 ns	

^ans = insignificant, * = significant at $p > 0.1$, ** = highly significant.

baseline practices showed pronounced variations in emission rates across sites and seasons.

The impact of the diverse mitigation practices on CH₄ emissions and rice yields are given in Tables 1 to 4 for each baseline practice separately. Results are specifically given for seasons because emissions showed pronounced interseasonal variations. This can be illustrated by the results from Maligaya where CH₄ emissions were generally high in the wet and low in the dry season (Figure 2). This interseasonal shift appears to be an influence of solar radiation and temperature difference during the early crop growth (Corton et al., this issue). Furthermore, these seasons also showed differences in the impact of mitigation practices. For example, ammonium sulfate was an efficient tool for mitigating emissions in the wet season, whereas its impact was insignificant in the dry season (Figure 2).

Baseline practice I_1 (continuous flooding, organic amendments)

The results listed in Table 1 underscore the potential of water regime to reduce CH₄ emissions from irrigated rice. Midseason drainage consistently lowered emissions as compared with a baseline practice of continuous flooding, but the reducing effect varied with both station and season. The reduction in emissions was statistically significant in seven out of the eight experiments; only the 1997 WS experiment at Maligaya showed an insignificant effect (Table 1). The reduction effect was highest when midseason drainage was supplemented by replacing organic manure with urea (Beijing, 1995 SS).

The practice of alternate flooding and drying of the field reduced emissions by about 60% as compared

Table 2. Mitigation practices for baseline practice I_2 (midseason drainage; organic amendments); see Table 1 for explanations

Mitigation practice	Seasonal emission (kg ha ⁻¹)	Reduction effect		Experiment		Yield impact (%)
		Net reduction (kg ha ⁻¹)	Relative reduction (%)	Station	Season	
Alternate flooding/drainage	207	178	46**	Beijing	1995 SS	-1.6
	217	95	30**	Hangzhou	1995 SS	2.8
	177	48	21**	Hangzhou	1998 ES	0.5
	215	64	23**	Hangzhou	1998 LS	4.2
Early single drainage	15	13	46**	Beijing	1998 SS	-1.7
Early dual drainage	21	7	25 ns	Beijing	1998 SS	0.3
Late dual drainage	26	2	7 ns	Beijing	1998 SS	1.2
Biogas manure	53	6	10**	Hangzhou	1997 ES	3.0
	151	28	16**	Hangzhou	1997 LS	-0.6
Rice straw winter application	200	25	11**	Hangzhou	1998 ES	-0.5
Mulching rice straw	248	31	11**	Hangzhou	1998 LS	0.0

with continuous flooding (Table 1). Alternate flooding/drying as well as midseason drainage had either no significant impact on yields or even increased rice production as compared with continuous flooding. However, temporary soil aeration could enhance emission of nitrous oxide (N_2O) (Bronson et al., 1997a,b; Abao et al., this issue), another potent greenhouse gas. When baseline levels of CH_4 emissions are high (as for I_1), midseason drainage can still be regarded as an efficient mitigation option in spite of concomitant increments in N_2O emissions (see discussion below).

The impact mechanisms of temporary soil aeration on CH_4 emissions have been discussed thoroughly in volume 2 of this series (Wassmann et al., this issue, c) as well as in various station reports (Wassmann et al., this issue, a; Lu et al., this issue; Wang et al., this issue). One rice crop requires about 1,240 mm water (Yoshida, 1981). To meet this water demand, wetland rice has evolved over the centuries as a well-adapted cultivation technique sustaining high yields. As opposed to upland systems, wetland cultivation also provides numerous advantages in terms of soil chemistry and erosion control. Large portions of Asia's rice land are on native wetlands that do not allow cultivating any crop other than rice (at least in the wet season).

The CH_4 -reducing effect of midseason drainage could substantially be enhanced in conjunction with direct wet seeding (Table 1). The practice of direct wet seeding alone accounted for a reduction effect of 16–22% in the seasonal emissions as compared with the baseline practice of transplanting. In Maligaya, the difference between direct-seeded and transplanted rice occurs relatively early in the season (Figure 3). Direct-

seeded rice develops high root biomass during early stages and reaches maximum root biomass soon after panicle initiation (De Datta & Nantasomsaran, 1991). Roots of transplanted rice develop slower but can penetrate into the deeper layer of the puddled soil as compared with the relatively compact soil under direct wet seeding. However, the precise mechanism involved in reducing CH_4 emissions through direct wet seeding still has to be clarified. Direct wet seeding is getting increasingly popular in major rice-growing regions. Substantial savings in labor requirements make this type of crop establishment economically viable, although yields are lower.

Application of rice straw compost significantly reduced CH_4 emissions as opposed to fresh rice straw (Table 1). The straw in the experiment in Maligaya was processed in an aerobic composter (Corton et al., this issue) so that CH_4 emissions during the composting process can be neglected. Addition of phosphogypsum also reduced CH_4 emissions. Phosphogypsum is a waste byproduct from processing of phosphate rock fertilizer and consists mainly of calcium sulfate dihydrate. The reduction effect of compost can be explained by a depletion of methanogenic substrate, whereas phosphogypsum triggers inhibition of methanogenesis through sulfate-reducing bacteria (Corton et al., this issue).

Baseline practice I_2 (midseason drainage, organic amendments)

In many rice-growing regions, the local practice of irrigation encompasses a drainage period in the early or midseason. While this drainage period itself exerts a

Table 3. Mitigation practices for baseline practice I₃ (continuous flooding, no organic amendments); see Table 1 for explanations

Mitigation practice	Seasonal emission (kg ha ⁻¹)	Reduction effect ^a		Experiment		Yield impact (%)	Remark
		Net reduction (kg ha ⁻¹)	Relative reduction (%)	Station	Season		
Preharvest drainage	251	(26)	(12*)	Los Baños	1994 DS	-6 ns	
	10	0	0 ns	Los Baños	1996 DS	-13 ns	
	28	12	30**	Los Baños	1996 WS	23 ns	
Dual drainage at midtillering and preharvest	45	180	80**	Los Baños	1994 DS	-4 ns	
	11	16	59**	Los Baños	1994 WS	-11 ns	
	8	2	20**	Los Baños	1996 DS	-9 ns	
	34	6	15**	Los Baños	1996 WS	-1 ns	
Alternate flooding/drying	14	4	22 ns	New Delhi	1997 WS	-9 ns	
Direct wet seeding	25	(10)	(67)	Thailand	1998 DS	20	
	25	(10)	(67)	Thailand	1998 DS	20	
	256	(27)	(12**)	Jakenan	1993 WS	59*	
Ammonium sulfate	230	36	14**	Maligaya	1994 WS	-2 ns	Pure ammonium sulfate in Maligaya;
	184	20	10**	Maligaya	1995 DS	-2 ns	
	327	191	37**	Maligaya	1995 WS	13 ns	Ammonium sulfate
	9	18	67**	Los Baños	1997 DS	-15 ns	Blended with urea
	7	6	46**	Los Banos	1997 WS	16 ns	
Tablet urea	104	66	39**	Jakenan	1996 WS	1 ns	
	163	18	10 ns	Jakenan	1997 DS	8 ns	
Phosphogypsum	145	83	36**	Maligaya	1995 DS	-2 ns	
	225	41	15**	Maligaya	1994 WS	-6 ns	
	241	25	9**	Maligaya	1994 WS	1 ns	
	143	388	73**	Maligaya	1995 WS	13 ns	
Rice stubbles and roots removed	14	26	65**	Los Baños	1996 WS	19 ns	
High-yielding variety (IR64)	115	26	18	Jakenan	1994 DS	10 ns	Baseline practice: IR72
New plant type	7	1	13**	Los Baños	1995 DS	-28 ns	Baseline practice: IR72
	6	2	25**	Los Baños	1995 WS	-50*	
Traditional variety	5	3	38*	Los Baños	1995 DS	-25 ns	Baseline practice: IR72
Hybrid rice	6	2	25*	Los Baños	1995 WS	0 ns	Baseline practice: IR72
	4	4	50 ns	Los Baños	1995 WS	69*	
	1	0	(12 ns)	Los Baños	1998 DS	-2 ns	
	1	0	(3 ns)	Los Baños	1998 DS	15*	
Direct wet seeding and midseason drainage	15.96	74	82	Maligaya	1998 DS	-4 ns	
Direct wet seeding and alternate flooding/drying	6.84	83	92	Maligaya	1998 DS	-11*	

^aFigures in brackets indicate a net increase in emission

Table 4. Mitigation practices for baseline practice R (rainfed rice) and baseline practice D (deepwater rice); see Table 1 for explanations

Mitigation practice	Seasonal emission (kg ha ⁻¹)	Reduction effect		Experiment		Yield impact (%)	Remark
		Net reduction (kg ha ⁻¹)	Relative reduction (%)	Station	Season		
<i>Baseline practice R</i>							
Farmyard manure	56	22	28**	Jakenan	1995 WS		Baseline practice
	92	(19)	(26**)	Jakenan	1996 DS		Rice straw
Compost	65	67	51**	Cuttack	1996 WS	-15*	Baseline practice: Sesbania
Azolla	68	64	48**	Cuttack	1996 WS	4 ns	Baseline practice: Sesbania
Nitrate inhibitor (Nimin)	77	(7)	(10)	Cuttack	1997 WS	24*	
Nitrate inhibitor (DCN)	61	9	13	Cuttack	1997 WS	31*	
Tablet urea	45	(8)	(22**)	Jakenan	1996 WS	-2 ns	Baseline practice: prilled urea
	102	(14)	(16**)	Jakenan	1997 DS	13 ns	
<i>Baseline practice D</i>							
No mineral fertilizer	213	19	10**	Prachinburi	1994 WS	11 ns	Baseline practice: mineral fertilizer with burned ash
Mineral fertilizer	201	7	4 ns	Prachinburi	1994 WS	-3 ns	
	48	19	28**	Prachinburi	1995 WS	-7 ns	
Compost straw	145	(85)	(142**)	Prachinburi	1997 WS	-8	
No mineral fertilizer and RS	53	14	21 ns	Prachinburi	1995 WS	-42 ns	
Mineral fertilizer and mulching RS	127	(58)	(84**)	Prachinburi	1996 WS	-3 ns	
Mineral fertilizer and RS	619	(550)	(797**)	Prachinburi	1996 WS	24 ns	
No tillage with mulching RS	100	(40)	(67**)	Prachinburi	1997 WS	-9	

mitigation effect, CH₄ emissions may further be reduced by a number of modifications (Table 2). Alternate flooding and drying in weekly intervals decrease emissions by 21-46%. Shifting the drainage period to an early stage or adding a second drainage period can also mitigate emissions by 7-46%.

Emissions from baseline practice I₂ can also be reduced through the management of organic amendments. As compared with pig manure application, the use of cattle manure and biogas residues (both are prefermented materials) reduced emissions by 77 and 10%, respectively. Rice straw can be applied during the winter fallow or can be mulched on the soil surface; both practices reduced CH₄ emissions by 11% as compared with incorporation of fresh straw into the soil during harrowing.

Baseline practice I₃ (continuous flooding, no organic amendments)

All modifications of floodwater regime that were considered for baseline practice I₁ may also be applied for reducing emissions in this baseline practice. A late drain-

age period alone did not reduce emissions efficiently, whereas a preceding drainage in the early stage had a pronounced reduction effect of 15-80% (Table 3). Direct wet seeding, however, yielded a net increase for baseline practice I₃ (without organic amendment) while it had a reductive effect for baseline practice I₁ (with organic amendment). Direct-seeded rice has a higher total root biomass (De Datta & Nantasomsaran, 1991) and thus may introduce more organic material into the soil. This additional substrate for methanogenic bacteria is more significant in a soil environment with low organic inputs as opposed to a soil with high organic inputs. While this explanation may provide an initial clue for the observed discrepancies, a fully satisfactory hypothesis on the impact of direct wet seeding on CH₄ emissions is still not possible. The impact of direct wet seeding in combination with midseason drainage and alternate flooding/drying, respectively, may primarily be attributed to changes in water regime and, only to a lesser extent, to effects on plant growth.

Emissions from baseline practice I₃ could be reduced by 10-67% through application of ammonium sulfate as opposed to urea (Table 3). Competition be-

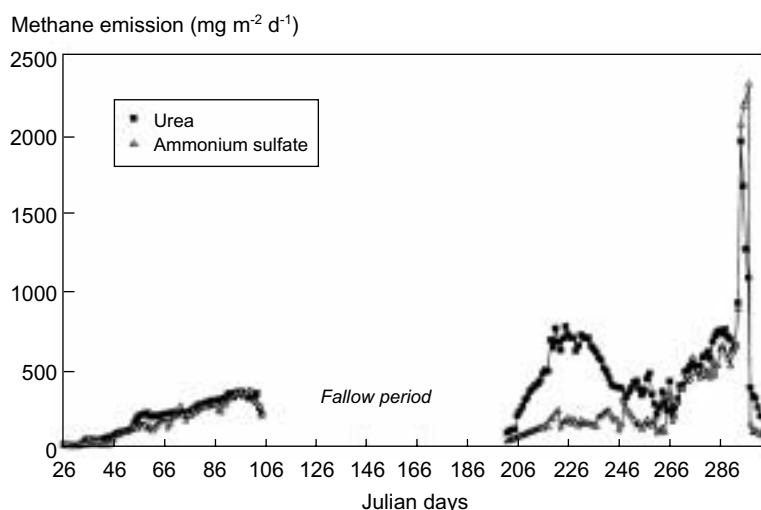


Figure 2. Methane emissions in 1995 dry and wet seasons (Maligaya) using different mineral fertilizers

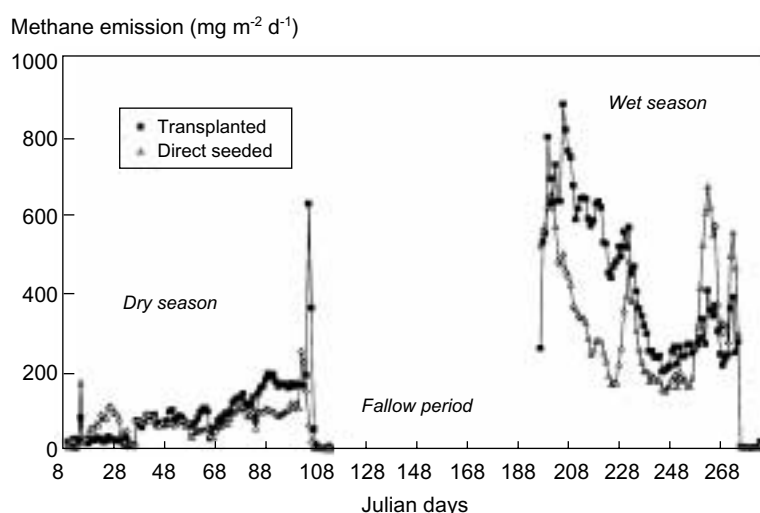


Figure 3. Methane emissions in 1997 dry and wet seasons (Maligaya) under different crop establishment methods

tween sulfate-reducing and methanogenic bacteria could be singled out to explain this effect (Corton et al., this issue). Likewise, the reductive effect of phosphogypsum could be explained by the sulfate content of this material (Corton et al., this issue). Application of tablet urea (a method for minimizing N losses) reduced CH_4 emissions by 10-39%. The experiment with stubbles and roots removed underscores the significance of residue management for CH_4 emissions, but this cannot be translated into a feasible mitigation option in farmers' field. Cultivar selection may become a mitigation option in the future, but the available database is still inconclu-

sive to draw definite recommendations on suitable varieties.

Baseline practices R (rainfed rice) and D (deepwater rice)

Table 4 shows a limited number of CH_4 mitigation options for rainfed and deepwater rice. While farmyard manure and rice straw have a similar emission potential, compost and azolla are a preferable manure type as opposed to sesbania. Based on the available database, nitrification inhibitors and tablet urea did not rep-

resent efficient tools for mitigating CH₄ emissions in rainfed rice.

Deepwater rice also offers very limited management options to reduce CH₄ emissions. Plant biomass is substantially higher than in irrigated and rainfed rice, so that straw management plays a crucial role in controlling emissions. The common practice of straw burning reduced CH₄ emissions during the ensuing growing season but contaminated the air with aerosols and gases including CH₄ during the burning process. Mulching of rice straw entails more emissions than ash application but less emissions than direct incorporation of fresh straw. This technique may still be recommended as a preferable straw management practice in deepwater rice.

Interaction with productivity of rice systems

Productivity of rice systems can be expressed as total factor productivity (grain output divided by all inputs taken together) (Dawe & Dobermann, 1999). While most of the suggested mitigation options did not affect yields significantly (Table 1-4), their profitability is primarily determined by fertilizer and labor inputs. Economically sound doses of fertilizers may also be beneficial for the greenhouse gas budget because (i) excessive supply of N entails N₂O emissions (Smith et al., 1997; Freney, 1997) and (ii) deficiencies of nutrients such as phosphorus increases root exudation and subsequently CH₄ emissions (Lu et al., 1999).

Replacing urea with ammonium sulfate, however, may be limited by economic disadvantages. Ammonium sulfate is more expensive than urea based on N content—i.e., the costs are about twice as high in the Philippines (FADINAP, 1999). On the other hand, ammonium sulfate has gained some acceptance as fertilizer (its consumption in Indonesia corresponds to 13% of urea consumption) (FADINAP, 1999) due to easy handling, storage, and application qualities. In rice production, ammonium sulfate is used mostly in the seedbed. Phosphogypsum is a byproduct of phosphate fertilizer manufacture; distinct opportunities for this soil additive for reducing CH₄ emission can be seen in sulfur-deficient soils.

The mitigation options addressing straw management and crop establishment entail changes in labor inputs. Farmers generally prefer removal of straw from the rice land because it can exacerbate soil tillage if present in large quantities. Composting of rice straw, on the other hand, represents additional work and limits farmer acceptance. However, the use of organic

amendments particularly rice straw is being promoted by most national extension services. Composting the rice straw may offer a number of benefits for soil fertility and tillage as opposed to fresh rice straw incorporation.

Biogas production could represent a low-cost source of energy for farmers, especially those with animals. In combination with rice production, biogas technology can achieve a twofold reduction of greenhouse gases (Wassmann et al., 1993): (1) prevention of CO₂ emissions by using renewable energy source and (2) reduction of CH₄ emissions from rice fields by replacing fresh manure with prefermented material. Biogas generation was successfully promoted in particular regions of China (e.g., Sichuan Province comprises about 7 million generators). Technical problems in operating small-scale biogas generators, however, have impeded their functionality and effectively stalled a further dissemination of these devices in rural areas (Wassmann et al., 1993). The beneficial greenhouse budget may become decisive arguments in favor of biogas generation in the future.

Direct wet seeding is an economically viable technique as opposed to the labor-intensive transplanting of rice plants although yields are lower. However, direct wet seeding is only recommended in systems with high organic inputs, which may also increase labor cost as compared with mineral fertilizer application.

The supply of water may incur costs for some rice farmers, e.g., through pumping. Moreover, water will become a scarce commodity in the future. Water-saving techniques can offer distinct trade-offs for mitigating CH₄ emissions as shown in this study for midseason drainage and alternate flooding/drying. While intermittent irrigation can substantially increase water use efficiency (Didiek, 1998), good timing of drainage and irrigation is essential to prevent soil compaction and subsequent water losses in reflooding the field (Tuong et al., 1996).

Midseason drainage is commonly practiced in Chinese rice fields as part of a high-yielding crop management, whereas Southeast Asian rice fields are generally not drained during the growing season. In the wet season, high precipitation constrains the effectiveness of field drainage. In the dry season, farmers are reluctant to remove water from the fields because of uncertain water supply for the remaining growing period. However, improved irrigation schemes could help in developing irrigation patterns that improve productivity and reduce CH₄ emissions.

Overall assessment of mitigation practices

The preferable mitigation options are listed in Table 5 for irrigated rice. While irrigated rice can be altered in virtually all aspects of crop management, mitigation options in rainfed and deepwater rice are very limited. However, Table 5 does not include (a) the use of chemical fertilizers, (b) straw burning, and (c) selection of rice cultivars as mitigation option for these reasons:

1. The possible mitigation effect of chemical fertilizers may be offset by CO₂ emissions through industrial N₂ fixation. One mole of ammonia fixed through the Haber-Bosch process produces 1.436 moles of CO₂ (Schlesinger, 1999). Application of 120 kg N ha⁻¹—as in the chemical fertilizer treatments of our experiments—translates into off-site emissions of 541.5 kg of CO₂. Using a conversion factor of 21 for the global warming potential of CH₄ in comparison with CO₂,

(IPCC 1995), this off-site emission of CO₂ corresponds to the radiative forcing of 25.8 kg CH₄ ha⁻¹.

2. Emissions of CH₄ resulting from rice straw burning are in the range of 0.43-0.90 % of the carbon content, which is similar to the range through straw application into the soil (Miura & Kanna, 1997). Moreover, straw burning emits significant quantities of other greenhouse gases such as CO and N₂O (Miura & Kanna, 1997) and adversely affects local air quality.
3. The database on rice cultivars affecting CH₄ emissions is still inconsistent. The two important traits that determine the CH₄ emission potential of rice cultivars are (a) root exudation and (b) gas transfer through the aerenchyma (Butterbach-Bahl et al., 1997). However, the CH₄ emission potential of a given cultivar exhibits enormous variation when grown under different greenhouse and field conditions (Wassmann & Aulakh, 1999). These variations

Table 5. Mitigation matrix for different baseline practices of irrigated rice; reduction effect for each mitigation practice is given in parentheses; arrows indicate that mitigation practice can be adopted to other baseline practices although experimental results are not available

Modified crop management	Baseline practice I ₁ (continuous flooding/ organic amendment)	Baseline practice I ₂ (midseason drainage, organic amendment)	Baseline practice I ₃ (continuous flooding, no organic amendment)
Water regime	Midseason drainage (7-44%) Alternate flooding/ drying (59-61%)	Alternate flooding/ drying (21-46%) Early/ dual drainage (7-46 %)	Midseason drainage (15-80%) Alternate flooding/ drying (22%)
Organic amendments	Compost (58-63%)	Biogas residues (10-16 %)	
Mineral amendments	Phosphogypsum (27-37%)		Phosphogypsum (9-73%) Ammonium sulfate (10-67%)
Straw management		Fallow incorporation (11%) Mulching (11%)	Tablet urea (10-39%)
Crop establishment	Direct wet seeding (16-22%)		

complicate determination of cultivar-specific emission potentials (Aulakh et al., 1999). Therefore, at this point, it is difficult to recommend preferable rice cultivars for mitigating CH₄ emissions. Nevertheless, selection of cultivars may become an important option in the future when information on the interaction of genotype and environment in determining the respective traits become available.

Modifications of irrigation patterns are only recommended when substantial amounts of organic material is used (Table 5). Modifications of the water regime are likely to affect emissions of other greenhouse gases from rice production, namely N₂O (Bronson et al., 1997a,b; Abao et al., this issue). This greenhouse gas contributing about 6% of the anthropogenic greenhouse effect (IPCC, 1996) is generated through nitrification and denitrification occurring in soils (Rennenberg et al., 1996). When rice fields are continuously flooded during the growing season, N₂O emissions are primarily limited to the fallow period at which fields experienced alternative dryness and wetness from rainfall (Abao et al., this issue). Water regimes that encompass drainage periods stimulate nitrification (through soil drying) and denitrification (through soil wetting). Therefore, all strategies to reduce CH₄ emissions by midseason or frequent drainage may enhance N₂O emissions. Based on a global warming potential of 310 for N₂O as opposed to 21 for CH₄ (IPCC, 1995), the observed net reductions of 118 kg CH₄ ha⁻¹ (Beijing) and 245 kg CH₄ ha⁻¹ (Hangzhou) with midseason drainage would theoretically be compensated for by concomitant increments in N₂O emissions of 8 and 16.5 kg N₂O ha⁻¹, respectively. However, total N₂O emissions under comparable flooding regimes as in these experiments were 1.6 kg N₂O ha⁻¹ in northeast China (Chen et al., 1997) and 2.4-6.2 kg N₂O ha⁻¹ in central China (Cai et al., 1997); the average for the entire country under different crop management practices is given at 2.4 kg N₂O ha⁻¹ (Xing & Xu, 1997). Thus, modification of water regime appears as a promising option to achieve net gains in greenhouse gas emissions when the baseline of CH₄ emissions is very high. In low CH₄-emitting rice systems, however, the net effect of modifying water regimes may in fact become negative in terms of radiative forcing of the gases emitted (Bronson et al., 1997b).

The overall aim cannot be to reduce CH₄ emissions to a zero level. A large portion of Asian rice fields are located on lowlands that would be flooded natu-

rally (at least for some time over the year). Natural wetlands are a source of CH₄, so the net effect of growing rice on this land is less than the actual emissions. Furthermore, CH₄ emission deriving from rice is only a small driver of global warming that is mainly caused by CO₂ emitted through combustion of fossil fuels. Even for CH₄ alone, the contribution of rice fields to the global CH₄ budget ranges from 2% to 5% (Matthews et al., this issue). On a national scale, however, rice is still the prevailing CH₄ source in most of Asia. In most countries of South, Southeast, and East Asia, emissions from rice fields are too high to be ignored as a possible avenue for reduction.

Recommendations

While this study has identified possible candidates for mitigating emissions, the successful implementation of different crop management practices for reducing emissions will depend on the outcome of future research. The following objectives must be targeted:

- Identifying high CH₄-emitting rice systems
- Characterizing site-specific settings for mitigation
- Developing packages of mitigation technologies on regional bases
- Ascertaining synergies with improving productivity
- Accounting for N₂O emissions

Identifying high CH₄-emitting rice systems

Given the spatial variability in emission rates, the most promising approach for effective mitigation is targeting those rice systems with high CH₄ emission potentials. Identification of high CH₄-emitting rice systems requires geographic data on rice ecosystems, crop management, soil, and climate. Ideally, these data should be incorporated in a geographic information system (GIS) that can be linked to a CH₄ model. An initial GIS database (Knox et al., this issue) and a CH₄ model (Matthews et al., this issue) have been developed as part of this project. The accuracy of both components, however, may be improved in the future. Incorporation of regional soil surveys in major rice-growing areas would substantially improve the accuracy of upscaling as compared with the use of global soil maps. Moreover, the current database covers only the five collaborating countries of the project and therefore should be extended to other rice-growing areas.

Characterizing site-specific conditions for mitigation

Methane emissions from rice fields cannot be reduced by using a 'blanket' strategy for the different rice-growing systems. The distinction of baseline scenarios (as done in the present study) is an initial step to classify rice systems for identifying adequate mitigation options. However, more site-specific information is needed to define the best strategy under certain natural and socioeconomic settings. Again, GIS databases could be deployed for such site characterizations.

Developing sound packages of mitigation technologies

Concepts for reducing CH₄ emissions have to consider intrinsic links between individual modifications—e.g., changes in straw treatment have to concur with appropriate soil tillage, timing of fertilizer application with irrigation pattern, etc. Furthermore, it seems unlikely that the optimum reduction effect can be accomplished by one modification only. Packages of technologies have to be based on a site-specific characterization as listed above.

Ascertaining synergies with improving productivity

As for other innovations, the success of mitigation strategies in farming practice will ultimately depend on their economic performance. Possible trade-offs between mitigation strategies and productivity could be derived from the following relationships:

- Methane and N₂O emissions represent major pathways for energy and nutrient losses, respectively, for the rice system.
- A balanced nutrient supply prevents excessive emissions related to phosphorus deficiencies and oversupply of N.
- Modern rice plants are characterized by low root exudation leading to relatively low CH₄ emission rates.
- Temporary soil aeration reduces CH₄ emission while yields may increase and water demand may decrease depending on plant and soil type, respectively.
- The increment in CH₄ emission rates triggered by organic material can greatly be reduced by biogas techniques and applying fermented (composted) crop residues, which in turn improves soil fertility.

Accounting for N₂O emissions

It is imperative to ensure a positive net balance in greenhouse gas emissions through recommended changes in crop management. The effects on N₂O have to be elucidated further for incorporation in a decision support system.

Conclusions

The largest share of historical and current greenhouse gas emissions has come from developed countries, but different countries have distinct capabilities for coping with climate change in the widest possible cooperation (Dixon et al., 1996). These principles were acknowledged in the Framework Convention on Climate Change. With the specifications of the Kyoto Protocol, agriculture research may in the future increasingly be concerned with greenhouse gas emissions and its prevention (Smith, 1999). In countries with predominant rice cultivation, rice research could play a crucial role in accomplishing the national goals stipulated in this convention.

The achieved outputs of the Interregional Program on Methane Emission from Ricefields have opened up the possibilities to immediately develop some specific mitigation technologies for defined target areas. However, the implementation of mitigation strategies has to be seen in the context of a socioeconomically sound rice production. Increasing rice production is imperative for future generations. The challenge for rice research is to develop technologies that increase rice yields and—at the same time—reduce greenhouse gas emissions.

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